

Milestone 3

CNN-Based Scanner Identification

Objective of Milestone 3

The objective of **Milestone 3** is to develop a **deep learning–based scanner identification system** using a **Convolutional Neural Network (CNN)**.

The model aims to automatically learn scanner-specific patterns directly from document images and accurately classify the scanner source.

This milestone extends the work of:

- **Milestone 1:** Image preprocessing and noise extraction
- **Milestone 2:** Hand-crafted feature–based machine learning models

by introducing **end-to-end learning using CNNs**.

Rational for Using CNN

Traditional machine learning models depend on manually designed features, which may not fully capture subtle scanner artifacts.

A CNN is chosen because it:

- Learns **spatial and texture-based features automatically**
- Captures **scanner-dependent noise and pixel inconsistencies**
- Eliminates the need for explicit feature engineering
- Provides higher robustness for image-based forensic analysis

Dataset Description

The dataset used in Milestone 3 is the **same labeled dataset generated in Milestone 1**, ensuring pipeline continuity. **Dataset File**

- `dataset_labels.csv`

Dataset Columns

Column	Description
scanner	Scanner category (class label)
original	Path to original scanned document
processed	Path to preprocessed grayscale image
noise	Path to extracted noise residual

Image Selection for CNN

- **Processed grayscale images** (processed column) were used as CNN inputs
- Images were resized to **128 × 128 pixels**
- Single-channel grayscale images reduce computational complexity while preserving scanner artifacts

Preprocessing Steps

Before CNN training, the following preprocessing steps were applied:

1. Load image paths from dataset_labels.csv
2. Read images in **grayscale mode**
3. Resize images to **128 × 128**
4. Normalize pixel values to the range **[0, 1]**
5. Encode scanner labels using **Label Encoding**
6. Convert labels to **one-hot encoded vectors**

CNN Architecture

The CNN was implemented using **TensorFlow (Keras API)**.

Architecture Overview

Layer	Purpose
Conv2D (32 filters)	Low-level feature extraction
MaxPooling2D	Spatial downsampling
Conv2D (64 filters)	Mid-level feature learning
MaxPooling2D	Dimensionality reduction
Conv2D (128 filters)	High-level feature extraction
MaxPooling2D	Feature compression
Flatten	Convert feature maps to vector
Dense (128 units)	Non-linear feature combination
Dropout (0.5)	Overfitting prevention
Dense (Softmax)	Multi-class classification

Trainable Parameters (Honest & Exact)

Total trainable parameters: 3,304,709

This relatively large parameter count allows the network to learn complex scanner-specific patterns effectively.

Model Training

Training Configuration

- **Optimizer:** Adam
- **Loss Function:** Categorical Cross-Entropy
- **Batch Size:** 32
- **Epochs:** 10
- **Train-Test Split:** 80% / 20%

Training and validation accuracy were monitored to ensure stable convergence and generalization.

Model Evaluation

The trained CNN was evaluated on the test dataset using standard classification metrics.

Metrics Used

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Performance Summary

- **Overall accuracy:** ~98%
- High precision and recall across most scanner classes
- Minimal confusion between scanner categories

Confusion Matrix Analysis

The confusion matrix shows **dominant diagonal values**, indicating:

- Correct classification for most samples
- Strong separation between scanner classes
- Very low misclassification rate

This confirms that the CNN successfully learned scanner-specific visual patterns.

Tools & Technologies Used

Tool	Purpose
Python	Core programming language
TensorFlow (Keras)	CNN implementation
NumPy	Numerical operations
Pandas	Dataset handling
Matplotlib	Visualization
Scikit-learn	Label encoding & evaluation
Visual Studio Code	Development environment

Conclusion

In Milestone 3, a **CNN-based scanner identification model** was successfully designed, trained, and evaluated.

The model demonstrated strong performance by learning discriminative scanner-specific features directly from document images, outperforming traditional feature-based approaches used in earlier milestones.

This milestone establishes a deep learning foundation for future work in document forensics and source attribution.